ECE 340: Semiconductor Devices

Lecture 24: P-N Junction Under Bias

Prof. Hyunseok Kim

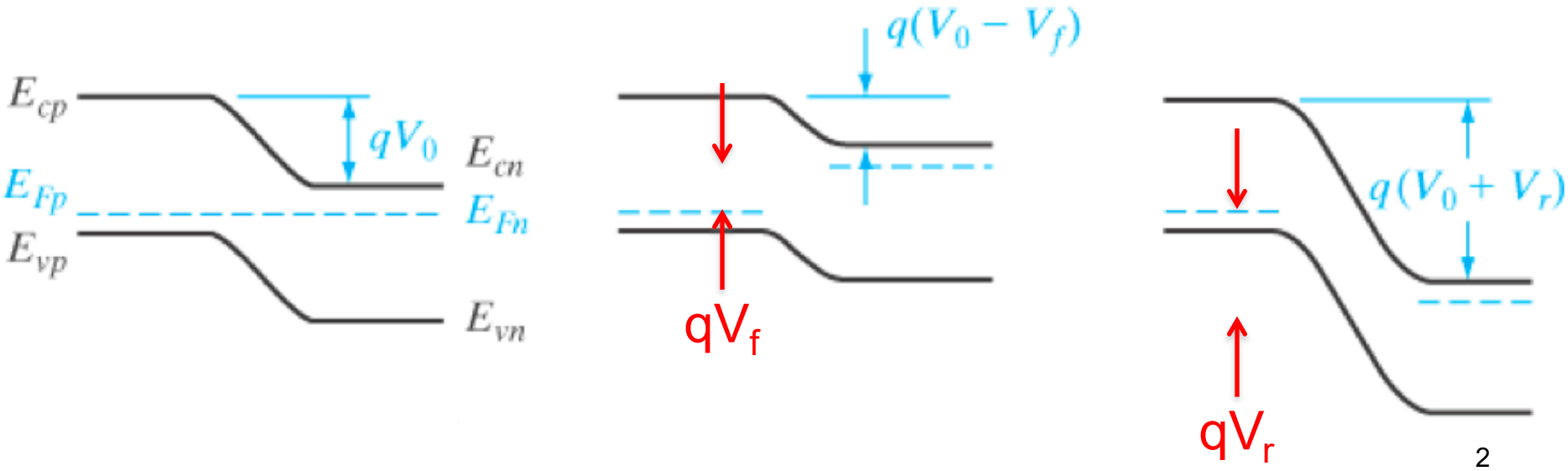
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Bias and Fermi Level Separation

- When we apply voltage we are no longer in equilibrium
- The application of a voltage creates a difference in the electrochemical potential across the junction boundary
- The Fermi levels become separated by an energy of "q" times the applied voltage
- The band separation is $q(V_0-V)$

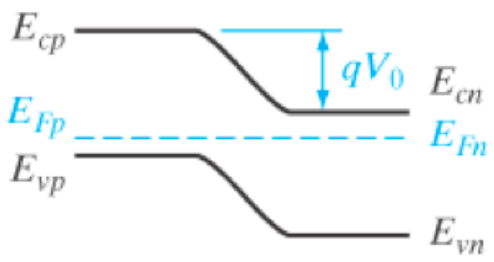


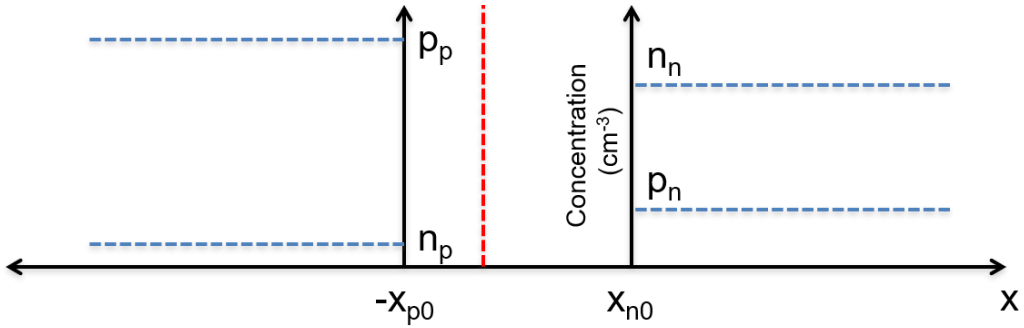
(Recap) Minority Carrier Concentration

Recall, under equilibrium

$$n(x) = N_C \exp\left[-\frac{E_C(x) - E_{Fn}}{kT}\right]$$

$$\frac{n_p(-x_p)}{n_n(x_n)} = \exp\left[-\frac{E_C(-x_p) - E_C(x_n)}{kT}\right]$$





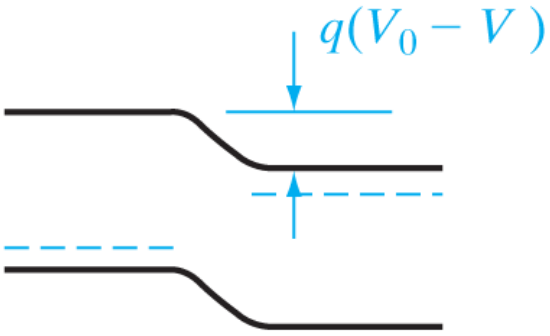
$$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$$

$$p_p = p_n e^{qV_o/kT} \quad \text{or} \quad p_n = p_p e^{-qV_o/kT}$$

$$n_p = n_n e^{-qV_o/kT} \quad \text{or} \quad n_n = n_p e^{qV_o/kT}$$

(Recap) Minority Carrier Concentration

Under bias (forward or reverse)



$$n(x) = N_C \exp\left[-\frac{E_C(x) - E_{Fn}}{kT}\right]$$

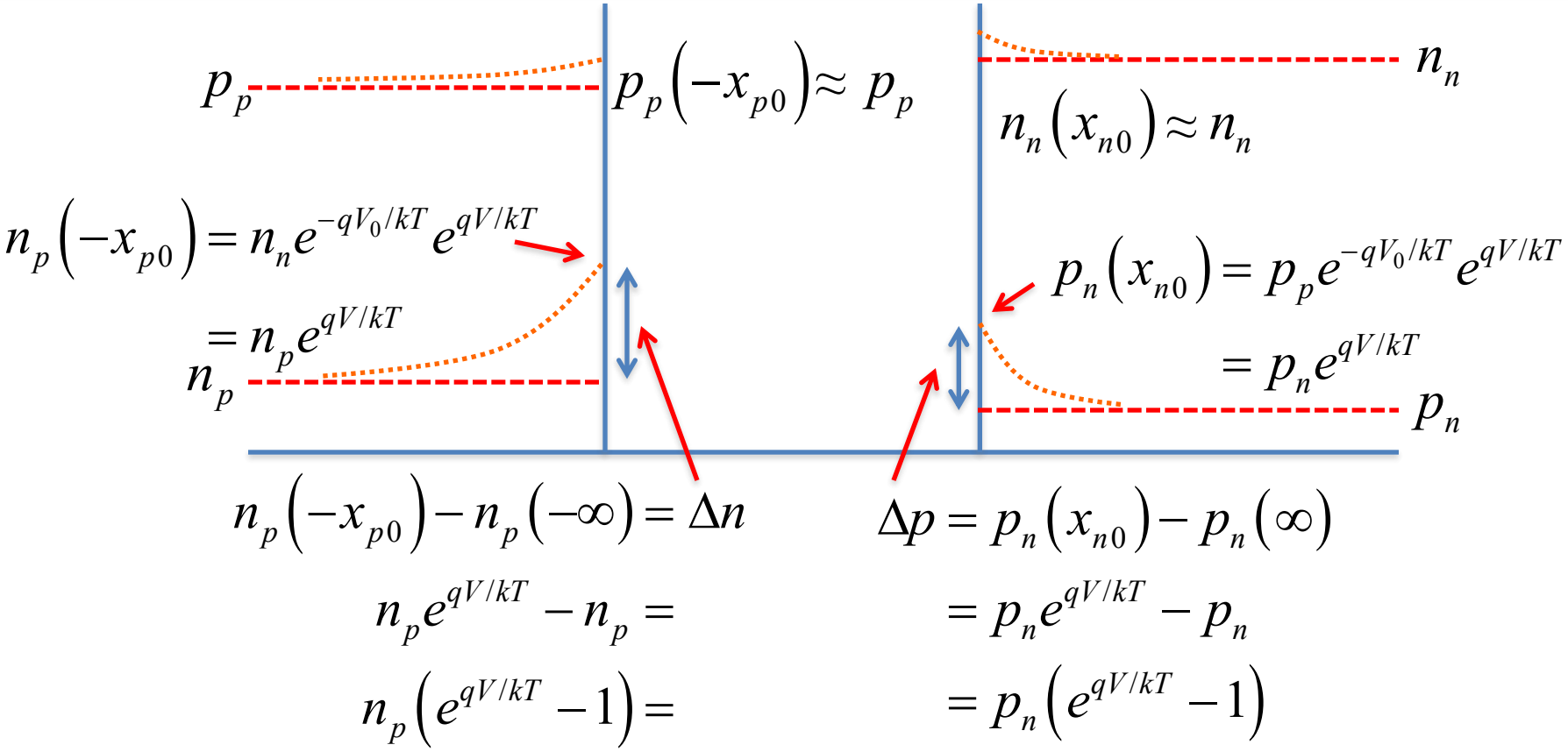
$$\frac{n_p(-x_p)}{n_n(x_n)} = \exp\left[-\frac{E_C(-x_p) - E_C(x_n)}{kT}\right]$$

Therefore,

$$n_p(-x_p) = n_n\, e^{-q(V_0 - V)/kT}$$

$$p_n(x_n) = p_p\, e^{-q(V_0 - V)/kT}$$

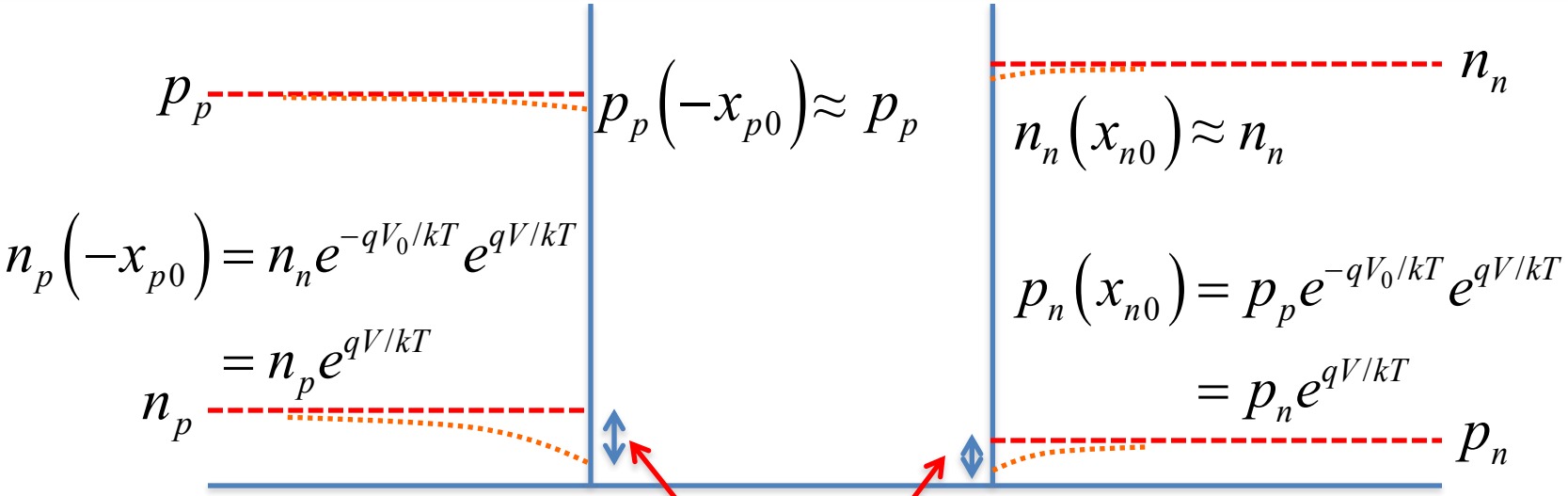
(Recap) Forward Bias



Space Charge Neutrality, Low - Level Injection :

$$p_p(-x_{p0}) = p_p(-\infty) + \Delta n = p_p + n_p(e^{qV/kT} - 1) \approx p_p$$

(Recap) Reverse Bias



$$n_p(-x_{p0}) - n_p(-\infty) = \Delta n$$

$$n_p e^{qV/kT} - n_p =$$

$$n_p\left(e^{qV/kT} - 1\right) =$$

$$-n_p \approx$$

$$\Delta p = p_n(x_{n0}) - p_n(\infty)$$

$$= p_n e^{qV/kT} - p_n$$

$$= p_n\left(e^{qV/kT} - 1\right)$$

$$\approx -p_n$$

Steady-State Carrier Injection and Diffusion Length

(Revisiting Earlier Material)

Diffusion Length: Steady State

Steady-State Diffusion Equations

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

In steady state, $\frac{\partial \delta p}{\partial t} = 0$

$$\therefore\ D_p \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{\tau_p} \text{ or } \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{D_p \tau_p} \equiv \frac{\delta p}{L_p^2}$$

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$

In steady state, $\frac{\partial \delta n}{\partial t} = 0$

$$\therefore\ D_n \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{\tau_n} \text{ or } \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2}$$

Hole Injection at x=0



(The injection mechanism is not specified.)

L_p is the average distance a hole travels before recombining.
L_n is the average distance an electron travels before recombining.

$$L_n \equiv \sqrt{D_n \tau_n} \text{ and } L_p \equiv \sqrt{D_p \tau_p}$$

Excess Hole Distribution and Current

Solution to Hole Diffusion Equation

$$\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2} \qquad \frac{\partial \delta p}{\partial t} = 0$$

Solution:

$$\delta p(x) = C_1 e^{x/L_p} + C_2 e^{-x/L_p}$$

Boundary Condition:

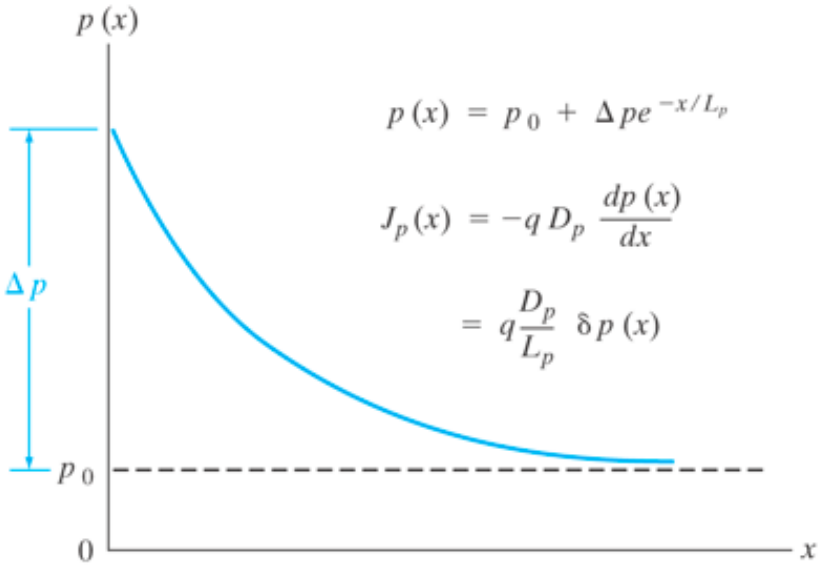
$$\delta p(x=0) = \Delta p = C_1 + C_2$$

$$\delta p(x=\infty) = 0 = C_1 e^{x/L_p}$$

$$C_1 = 0, \quad C_2 = \Delta p$$

$$\boxed{\delta p(x) = \Delta p \, e^{-x/L_p}}$$

Hole Injection at x=0



$$J_p(x) = -qD_p \frac{dp}{dx} = -qD_p \frac{d(p_o + \delta p)}{dx} = -qD_p \frac{d\delta p}{dx}$$

$$= q\frac{D_p}{L_p}\Delta p e^{-x/L_p} = q\frac{D_p}{L_p}\delta p(x)$$

Excess Electron Distribution and Current

Solution to Electron Diffusion Equation

$$\frac{d^2\delta n}{dx^2} = \frac{\delta n}{L_n^2} \qquad \frac{\partial \delta n}{\partial t} = 0$$

Solution:

$$\delta n(x) = C_1 e^{x/L_n} + C_2 e^{-x/L_n}$$

Boundary Condition:

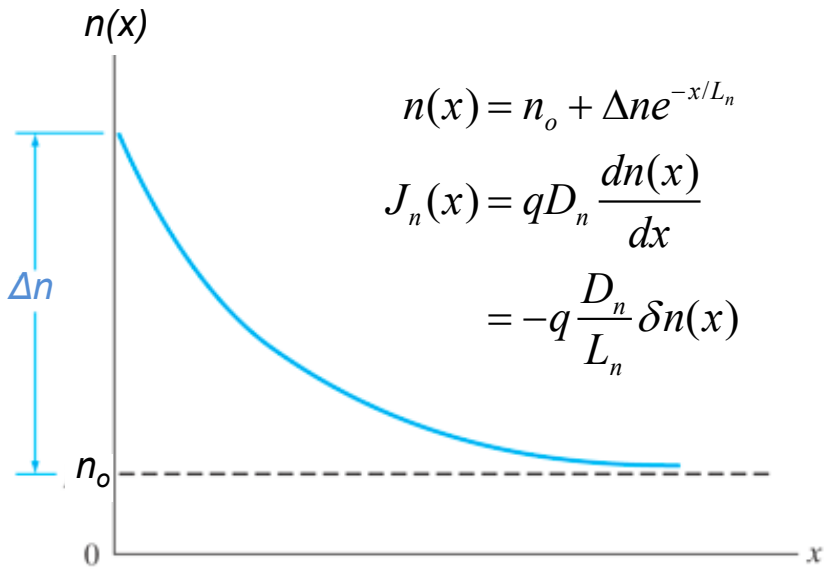
$$\delta n(x=0) = \Delta n = C_1 + C_2$$

$$\delta n(x=\infty) = 0 = C_1 e^{x/L_n}$$

$$C_1 = 0, \quad C_2 = \Delta n$$

$$\boxed{\delta n(x) = \Delta n\, e^{-x/L_n}}$$

Electron Injection at x=0



$$n(x) = n_o + \Delta n e^{-x/L_n}$$

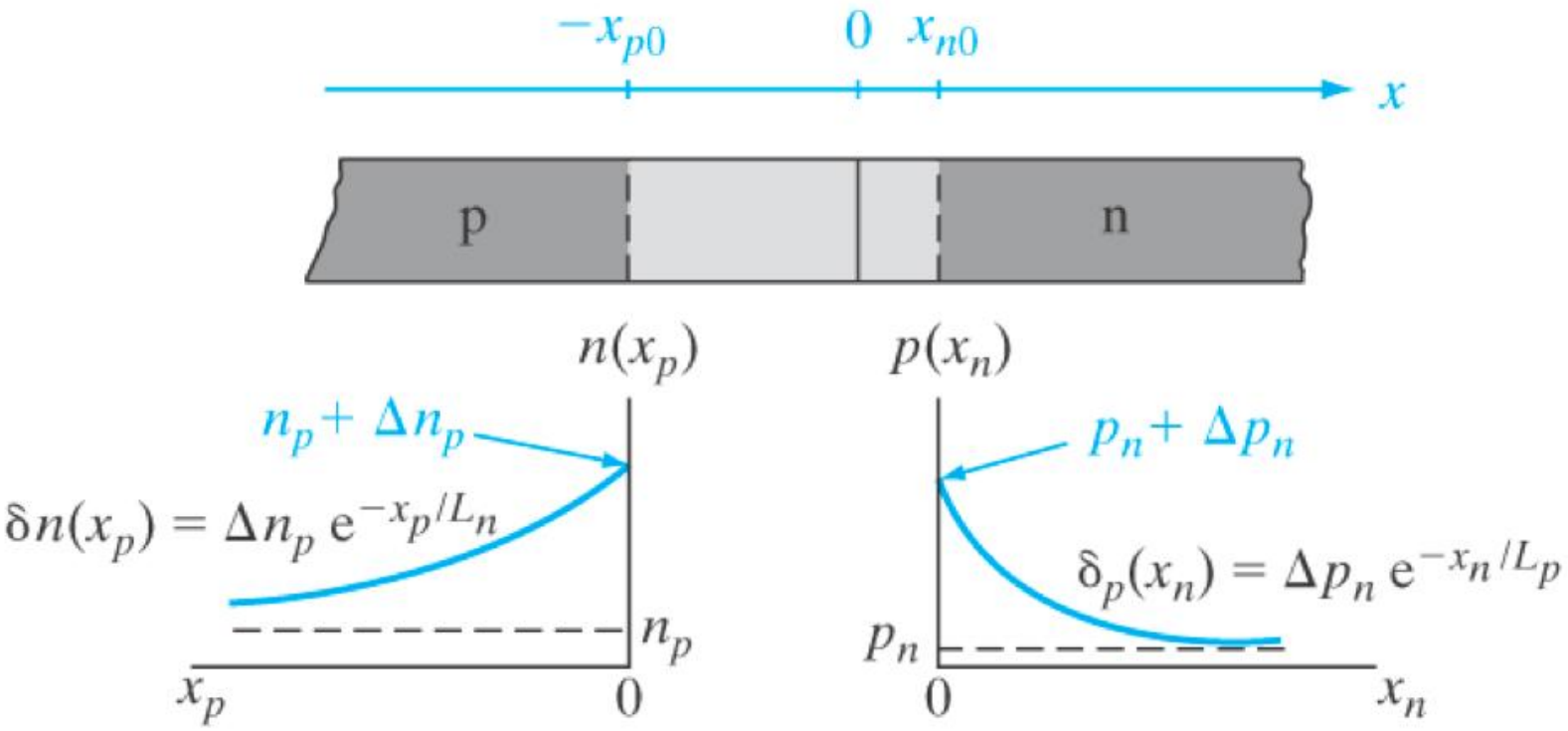
$$J_n(x) = qD_n \frac{dn(x)}{dx} = -q\frac{D_n}{L_n}\delta n(x)$$

$$\boxed{J_n(x)} = qD_n \frac{dn}{dx} = qD_n \frac{d(n_o + \delta n)}{dx} = qD_n \frac{d\delta n}{dx}$$

$$= -q\frac{D_n}{L_n}\Delta n e^{-x/L_n} = \boxed{-q\frac{D_n}{L_n}\delta n(x)}$$

Back to Carrier Injection

The Case We Need to Examine



Coordinate Transformation:

$x_n = x - x_{n0}$

$x_p = -x - x_{p0}$

x_{n0} and x_{p0} are the n and p-side depletion widths

Minority Carrier Distribution

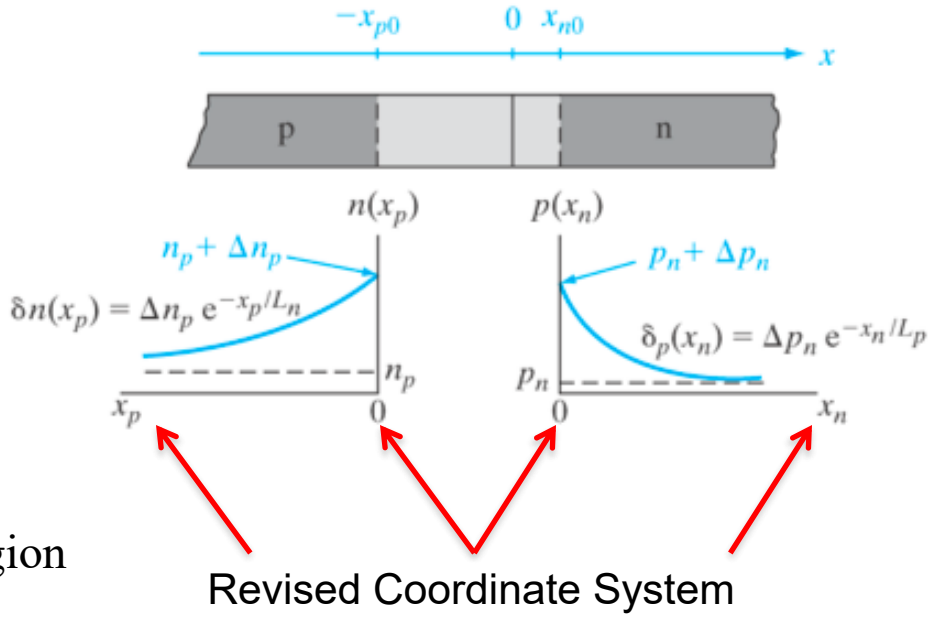
Define new coordinate systems referenced to edges the of the depletion region (x_{no} and $-x_{po}$). The diffusion equation for steady-state injection can be written:

$$\delta p(x_n) = \Delta p e^{-x_n/L_p} = p_n\left(e^{qV/kT} - 1\right)e^{-x_n/L_p}$$

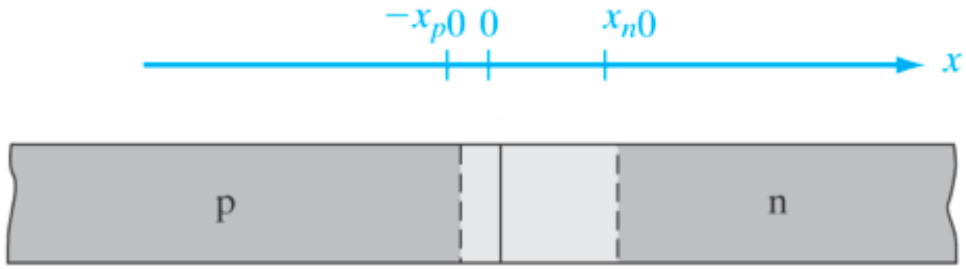
similarly, for electrons as the minority carrier:

$$\delta n(x_p) = \Delta n e^{-x_p/L_n} = n_p\left(e^{qV/kT} - 1\right)e^{-x_p/L_n}$$

where x_n is referenced to the edge of depletion region on the n-side and extends in the +x direction, and x_p is referenced to the edge of the depletion region on the p-side and extends in the -x direction



Total current density Jn+Jp should be constant



Recall continuity equation:

$$\frac{\partial n}{\partial t} = G_n - U_n + \frac{1}{q}\nabla \bullet \vec{J}_n$$

$$\frac{\partial p}{\partial t} = G_p - U_p - \frac{1}{q}\nabla \bullet \vec{J}_p$$

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} + G$$

$$\frac{\partial n(x,t)}{\partial t} = \frac{\partial \delta n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n} + G$$

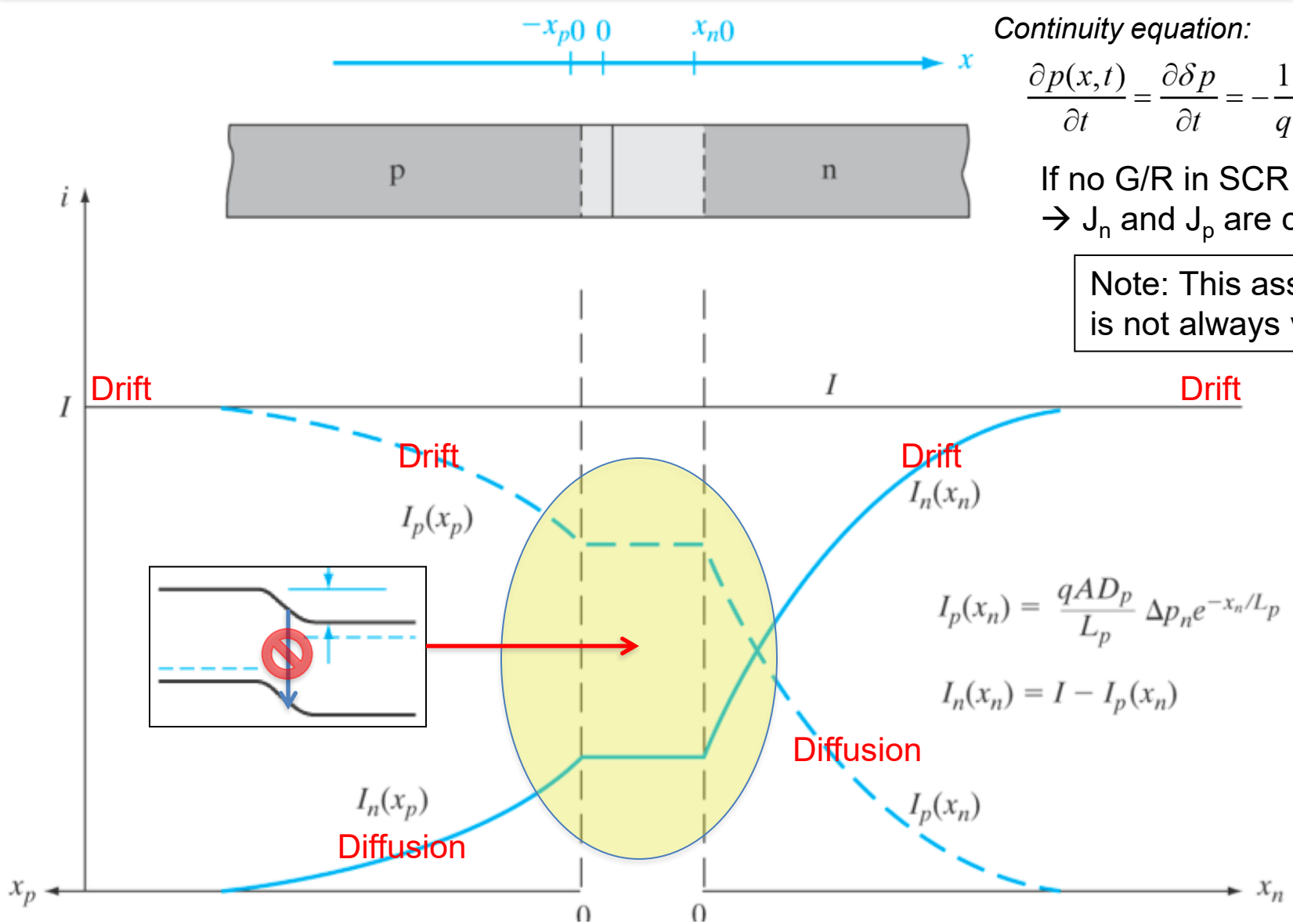
In steady state, $\frac{\partial p(x,t)}{\partial t} = 0$ $\frac{\partial n(x,t)}{\partial t} = 0$

$$\frac{\partial J_p}{\partial x} = q(R - G) \qquad \frac{\partial J_n}{\partial x} = -q(R - G)$$

$$\frac{d}{dx}(J_n + J_p) = 0$$

Therefore, J_n+J_p is constant throughout *x*. (But individual Jn or Jp may not be)

No Recombination in Depletion Region



Continuity equation:

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} + G$$

If no G/R in SCR region
→ J_n and J_p are constant!

Note: This assumption is not always valid.

$$I_p(x_n) = \frac{qAD_p}{L_p}\Delta p_n e^{-x_n/L_p}$$

$$I_n(x_n) = I - I_p(x_n)$$

The Ideal Diode Equation

Hole Current

Recall that:

$$J_p(x) = q\frac{D_p}{L_p}\delta p(x)$$

Therefore:

$$I_p(x_n = 0) = \frac{qAD_p}{L_p} p_n \left(e^{qV/kT} - 1\right)$$

Electron Current

Recall that:

$$J_n(x) = -q\frac{D_n}{L_n}\delta n(x)$$

Therefore:

$$I_n(x_p = 0) = -\frac{qAD_n}{L_n} n_p \left(e^{qV/kT} - 1\right)$$

** Relative to the transformed coordinate system**

Total Current

Neglecting recombination in the transition region:

$$\left|I_p(x_n = 0)\right| = \left|I_p(x_p = 0)\right| \text{ and } \left|I_n(x_p = 0)\right| = \left|I_n(x_n = 0)\right|$$

Since x_p is defined in the -x direction:

$$I_n(x_n = 0)_{x_n axis} = -I_n(x_p = 0)_{x_p axis}$$

The total current is the sum of the electron and hole currents:

$$I = I_p(x_n = 0) + I_n(x_n = 0)_{x_n axis} = I_p(x_n = 0) - I_n(x_p = 0)_{x_p axis} \text{ so:}$$

$$I = \frac{qAD_p}{L_p}\Delta p_n + \frac{qAD_n}{L_n}\Delta n_p = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT} - 1\right)$$

A Few Points

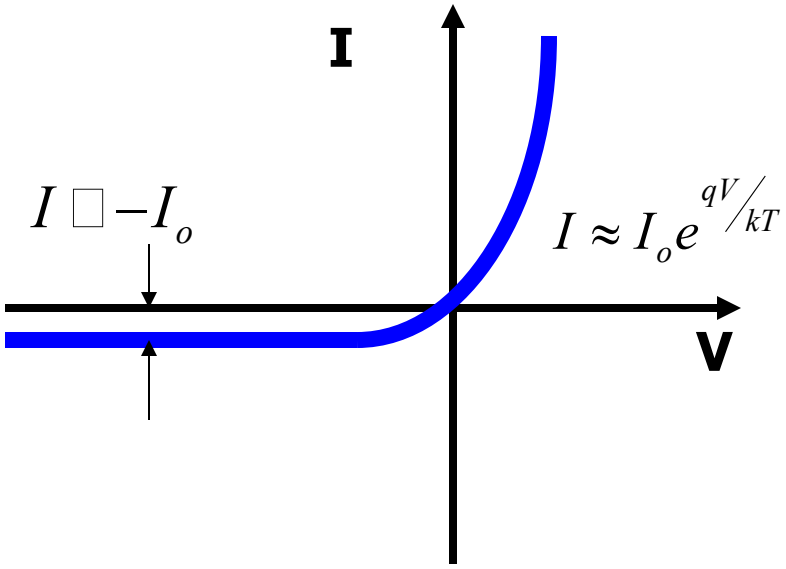
$$I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT} - 1\right) = I_o\left(e^{qV/kT} - 1\right)$$

n-side current (hole diffusion)

p-side current (electron diffusion)

- The dominant current contribution comes from injection from the more heavily doped side into the more lightly doped side
- **Minority carrier concentration**: Lower doping increases the minority carrier concentration (p_n or n_p), which increase the current for a given voltage.
- **Diffusion length:** In the meantime, higher doping decreases the minority carrier lifetime and (and thus also the diffusion length (L_p or L_n)), which increase the current for a given voltage.
- Thus, doping has competing effects on diode current, and the net effect depends on which contribution dominates.

Diode Equation: Various Conditions



$$I = I_o(e^{qV/kT} - 1)$$

$$I \approx I_o e^{qV/kT} \text{ for } V = V_f$$

$$I \approx -I_o \text{ for } V = -V_r$$

$$I = 0 \text{ for } V = 0$$

$$I = qA\left(\frac{D_p P_{on}}{L_p} + \frac{D_n n_{op}}{L_n}\right)(e^{qV/kT} - 1) = I_o(e^{qV/kT} - 1)$$

$(a)\ p - n\ Junction$

$$I = qA\left(\frac{D_p P_{on}}{L_p} + \frac{D_n n_{op}}{L_n}\right)(e^{qV/kT} - 1)$$

$(b)\ p^+ - n\ Junction$

$$p_{op} >> n_{on} \Rightarrow n_{op} << p_{on}$$

$$I = qA\left[\frac{D_p p_{on}}{L_p}\right](e^{qV/kT} - 1)$$

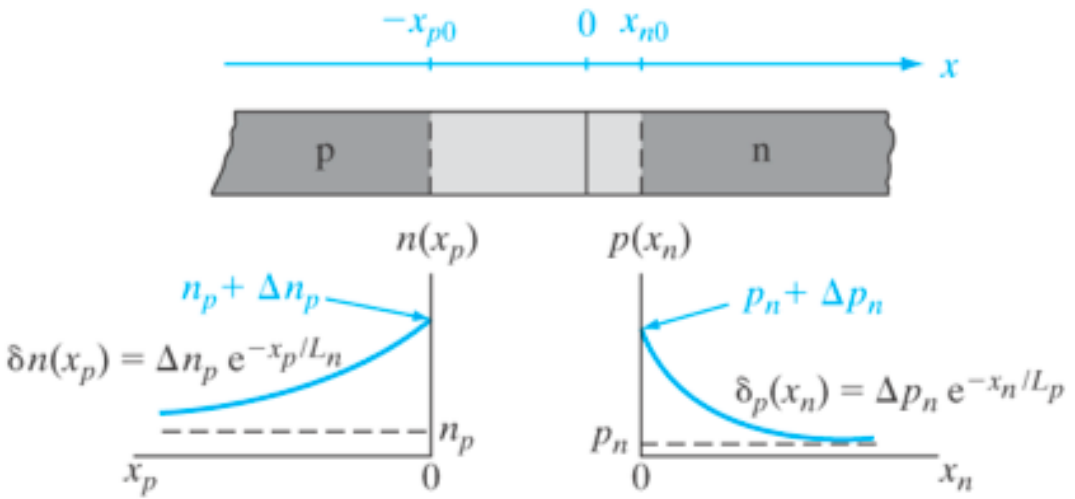
$(c)\ p - n^+\ Junction$

$$p_{op} << n_{on} \Rightarrow n_{op} >> p_{on}$$

$$I = qA\left[\frac{D_n n_{op}}{L_n}\right](e^{qV/kT} - 1)$$

Quasi Fermi Levels in the Forward Biased Junction

Quasi Fermi Levels in the Forward-Biased Junction



$$p(x_p = 0) \sim p_p(x_p = \infty) = p_p$$

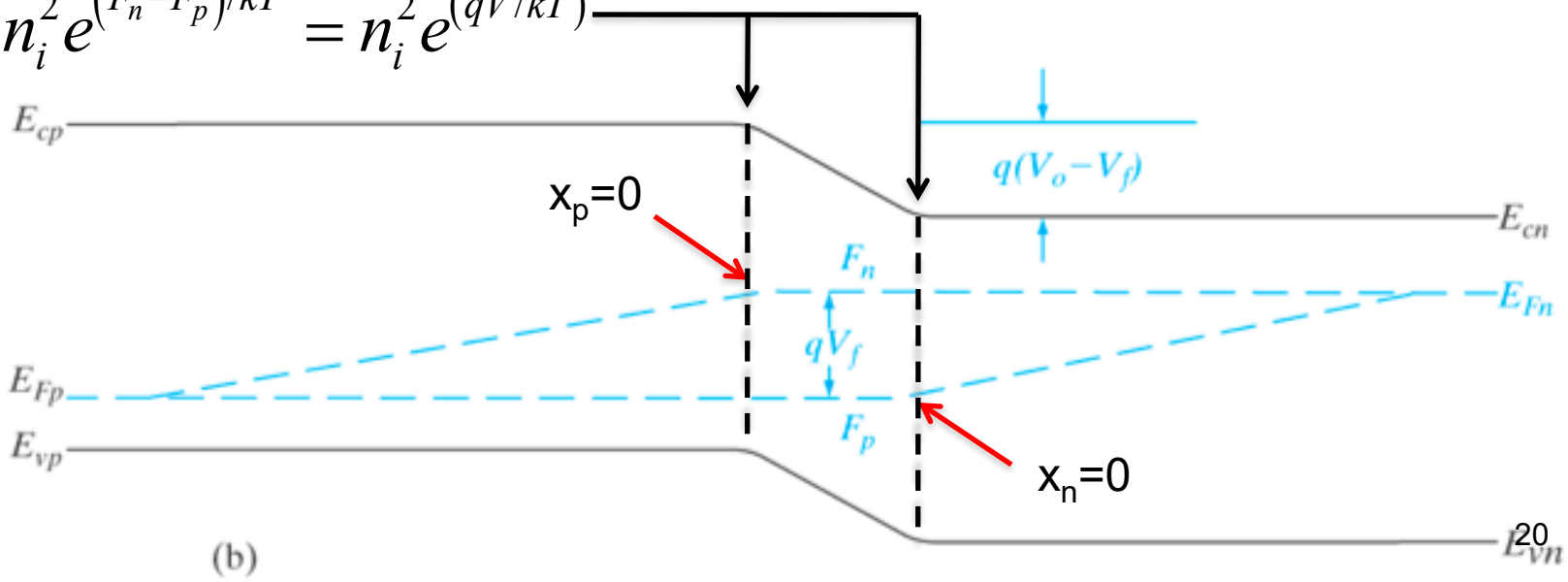
$$n(x_p = 0) = n_p(x_p = \infty) + \Delta n$$

$$= n_p\left(e^{qV/kT} - 1\right) \sim n_p e^{qV/kT}$$

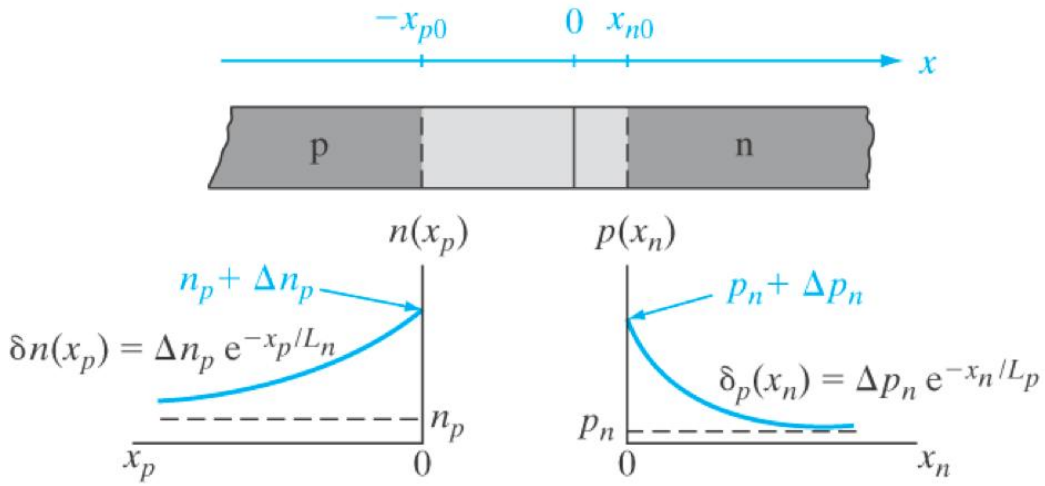
so:

$$pn(x_p = 0) = p_p n_p e^{qV/kT} = n_i^2 e^{qV/kT}$$

$$pn = n_i^2 e^{(F_n - F_p)/kT} = n_i^2 e^{(qV/kT)}$$



Relationship of pn to n_i^2 Under Bias



On the p-side of the junction:

$$p(x_p = 0) \sim p_p(x_p = \infty) = p_p$$

$$n(x_p = 0) = n_p(x_p = \infty) + \Delta n = n_p\left(e^{qV/kT} - 1\right) \sim n_p e^{qV/kT}$$

so:

$$pn(x_p = 0) = p_p n_p e^{qV/kT} = n_i^2 e^{qV/kT}$$

This expression applies at the edges of the depletion region

Majority and Minority Currents

Electron & Hole Currents

Ideal Diode Equation

$$I = \frac{qAD_p}{L_p}\Delta p_n + \frac{qAD_n}{L_n}\Delta n_p = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT} - 1\right)$$

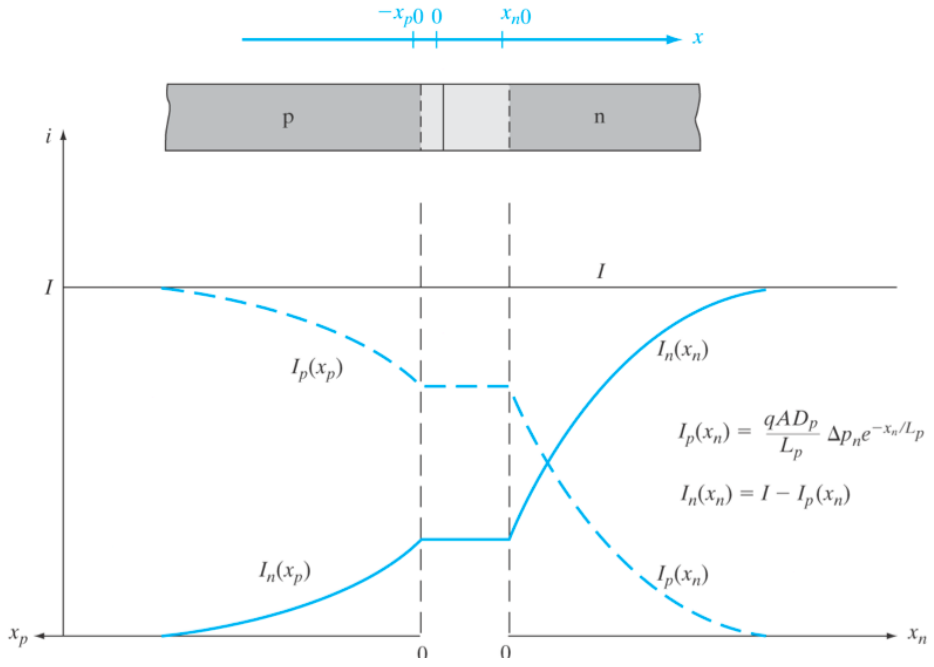
Defining $I_o = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)$, $I = I_o\left(e^{qV/kT} - 1\right)$

$$I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT} - 1\right) = I_o\left(e^{qV/kT} - 1\right)$$

Under large reverse bias $\left(V_r >> \frac{kT}{q}\right)$, $\left(e^{-qV_r/kT} - 1\right) \approx -1$ so

$$I = -qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right) = -I_o$$

Current Components



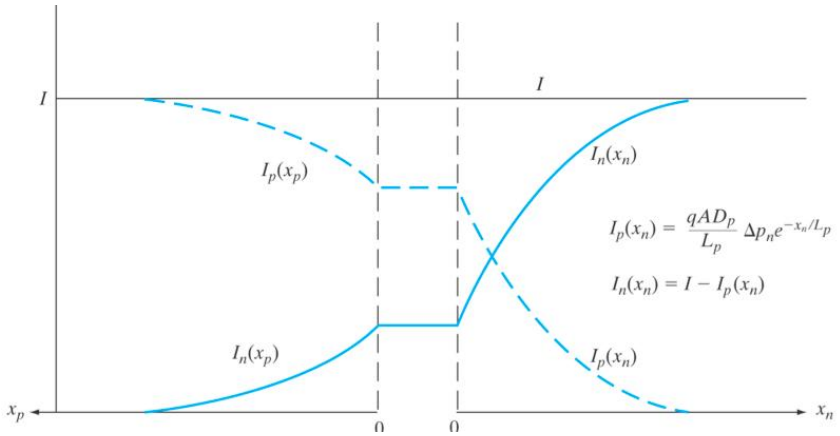
Majority Carrier Current: n-Side

Calculation of Majority Carrier Current :

$$I_{tot} = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT} - 1\right)$$

$$= I_{n_n} + I_{p_n} = I_n(x_n) + I_p(x_n) \quad \Leftarrow \quad \text{majority carrier + minority carrier current}$$

$$= I_n(x_n) + qA\frac{D_p}{L_p}p_n e^{-x_n/L_p}\left(e^{qV/kT} - 1\right)$$



so:

$I_{n_n} \equiv$ Majority Carrier Current

$I_{p_n} \equiv$ Minority Carrier Current

$I_n(x_n) \equiv$ Majority Carrier Current Referenced to Depletion Region Edge

$I_p(x_n) \equiv$ Minority Carrier Current Referenced to Depletion Region Edge

Majority Carrier Current: n-Side

$$I_{tot} = I_n(x_n) + qA\frac{D_p}{L_p}p_n e^{-x_n/L_p}\left(e^{qV/kT}-1\right)$$

so:

$$I_n(x_n) = I_{tot} - I_{p_n} = \left[qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT}-1\right)\right] - \left[qA\frac{D_p}{L_p}p_n e^{-x_n/L_p}\left(e^{qV/kT}-1\right)\right]$$

$$= qA\left[\frac{D_p}{L_p}\left(1-e^{-x_n/L_p}\right)p_n + \frac{D_n}{L_n}n_p\right]\left(e^{qV/kT}-1\right)$$

where x_n is the coordinate system with $x_n = 0$ at the edge of the n-side depletion region

Relative to the "x" coordinate system with the origin at the metallurgical junction:

$$I_n(x) = qA\left[\frac{D_p}{L_p}\left(1-e^{-(x-\bar{x}_n)/L_p}\right)p_n + \frac{D_n}{L_n}n_p\right]\left(e^{qV/kT}-1\right)$$

where $\bar{x}_n$ is the width of the depletion region on the n-side of the junction

Majority Carrier Current: p-Side

Calculation of Majority Carrier Current :

$$I_{tot} = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)\left(e^{qV/kT} - 1\right)$$

$$= I_{p_p} + I_{n_p} = I_n(x_p) + I_p(x_p) \quad \Leftarrow \text{ majority carrier + minority carrier current}$$

$$= I_p(x_p) + qA\frac{D_n}{L_n}n_p e^{-x_p/L_n}\left(e^{qV/kT} - 1\right)$$

so:

$I_{p_p} \equiv$ Majority Carrier Current

$I_{n_p} \equiv$ Minority Carrier Current

$I_p(x_p) \equiv$ Majority Carrier Current Referenced to Depletion Region Edge

$I_n(x_p) \equiv$ Minority Carrier Current Referenced to Depletion Region Edge

x_p is a transformed coordinate system: $x_p = -x - \overline{x}_p$

where:

The "x" coordinate system is the + going system referenced to the metallurgical junction

$\overline{x}_p$ is the p-side depletion region width

Majority Carrier Current: p-Side

$$I_{tot} = I_p(x_p) + qA\frac{D_n}{L_n} n_p e^{-x_p/L_n}\left(e^{qV/kT} - 1\right)$$

so:

$$I_p(x_p) = I_{tot} - I_{n_p} = \left[qA\left(\frac{D_p}{L_p} p_n + \frac{D_n}{L_n} n_p\right)\left(e^{qV/kT} - 1\right)\right] - \left[qA\frac{D_n}{L_n} n_p e^{-x_p/L_n}\left(e^{qV/kT} - 1\right)\right]$$

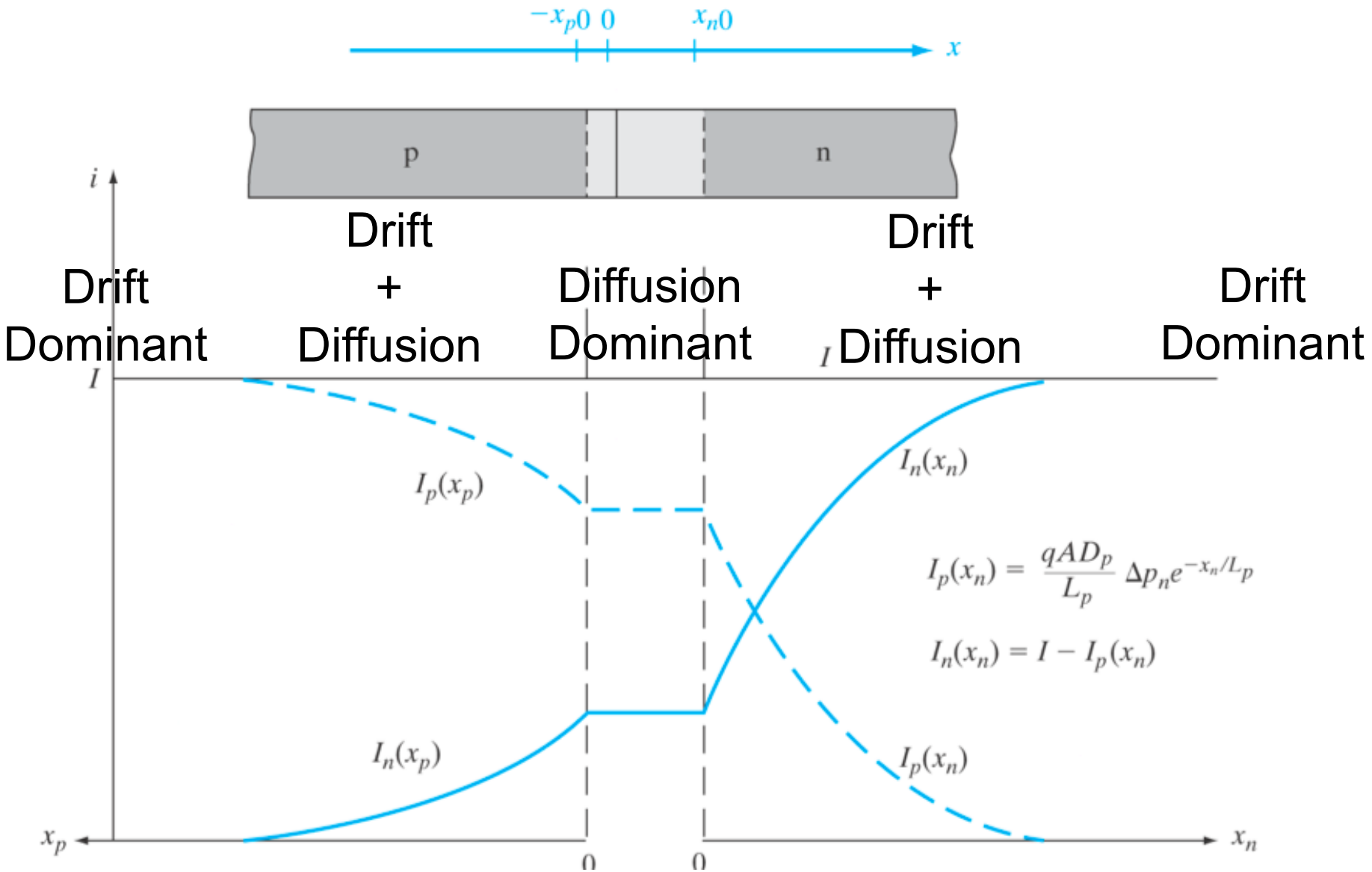
$$= qA\left[\frac{D_n}{L_n}\left(1 - e^{-x_p/L_p}\right) n_p + \frac{D_p}{L_p} p_n\right]\left(e^{qV/kT} - 1\right)$$

where x_p is the negative-going coordinate system with $x_p = 0$ at the edge of the n-side depletion region

Referenced to Metallurgical Junction and + going System:

$$I_p(x_p) = qA\left[\frac{D_n}{L_n}\left(1 - e^{x+\bar{x}_p/L_p}\right) n_p + \frac{D_p}{L_p} p_n\right]\left(e^{qV/kT} - 1\right)$$

Drift & Diffusion in Forward Bias



Example

Example: p-n Diode

Determine the current at bias voltages of +0.5 and -0.5V.

Silicon p-n junction with an area of 10^{-4} cm^2 at 300K:

P-N properties:

p-side	n-side
$N_a=10^{17}$ cm^{-3}	$N_d=10^{15}$ cm^{-3}
$\tau_n=0.1$ μs	$\tau_p=10$ μs
$\mu_p=200$ $cm^2/V\text{-}s$	$\mu_n=1300$ $cm^2/V\text{-}s$
$\mu_n=700$ $cm^2/V\text{-}s$	$\mu_p=450$ $cm^2/V\text{-}s$

Calculate equilibrium minority carrier concentrations:

$$p_n = \frac{n_i^2}{n_n} = \frac{\left(1.5\times10^{10}\right)^2}{10^{15}} = 2.25\times10^5\, cm^{-3}$$

$$n_p = \frac{n_i^2}{p_p} = \frac{\left(1.5\times10^{10}\right)^2}{10^{17}} = 2.25\times10^3\, cm^{-3}$$

Calculate the **MINORITY CARRIER** diffusion coefficient and diffusion length:

$$D_p = \frac{kT}{q}\mu_p = 0.0259\times450 = 11.66\ \text{cm}^2/\text{s} \leftarrow \text{holes on n-side}$$

$$D_n = \frac{kT}{q}\mu_n = 0.0259\times700 = 18.13\ \text{cm}^2/\text{s} \leftarrow \text{electrons on p-side}$$

$$L_p = \sqrt{D_p\tau_p} = \sqrt{11.66\times\left(10\times10^{-6}\right)} = 1.08\times10^{-2}\ \text{cm}$$

$$L_n = \sqrt{D_n\tau_n} = \sqrt{18.13\times\left(0.1\times10^{-6}\right)} = 1.35\times10^{-3}\ \text{cm}$$

Example: p-n Diode

Determine the current at bias voltages of +0.5 and -0.5V.

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μ_p=200 cm^2/V-s	μ_n=1300 cm^2/V-s
μ_n=700 cm^2/V-s	μ_p=450 cm^2/V-s

Calculate I_o :

$$I_o = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)$$

$$= \left(1.6\times10^{-19}\right)\times\left(0.0001\right)\left(\frac{11.66}{0.0108}\left(2.25\times10^{5}\right)+\frac{18.13}{0.00135}\left(2.25\times10^{3}\right)\right)$$

$$= 4.370\times10^{-15}\,\text{A} = 4.370\ \text{fA}$$

Current at a voltage of +0.5 V: $I = I_o\left(e^{0.5/0.0259}-1\right)\approx 1.058\times10^{-6}\,A$

Current at a voltage of -0.5 V: $I \approx -I_o = -4.37\times10^{-15}\,A$

Key Point:

- Use the correct mobility in the calculation of D_p and D_n
- Minority carrier mobility